

Applications of vectors in geometry

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1 Preliminaries

Recall the *dot* or *scalar* product of two vectors $\vec{u} = (a_1, \dots, a_n), \vec{v} = (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$ is defined as

$$\vec{u} \cdot \vec{v} = a_1 b_1 + \dots + a_n b_n \in \mathbb{R}.$$

The dot product has the following properties, which often reduces problems involving quadratic metric relations, perpendicularity, etc to algebraic computations.

1. (Symmetric bilinearity) $\vec{u} \cdot (\alpha \vec{v} + \beta \vec{w}) = \alpha \vec{u} \cdot \vec{v} + \beta \vec{u} \cdot \vec{w}, \forall \alpha, \beta \in \mathbb{R}.$
2. $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cos \angle(\vec{u}, \vec{v})$. In particular, $\vec{u} \perp \vec{v} \iff \vec{u} \cdot \vec{v} = 0.$
3. (Cauchy-Schwarz) $|\vec{u} \cdot \vec{v}| \leq |\vec{u}| \cdot |\vec{v}|$, with equality iff $\vec{u}, \vec{v}$ are collinear.
4. (Triangle inequality) $|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|.$

The *cross* product of two vectors $\vec{u}, \vec{v} \in \mathbb{R}^3$, denoted by $\vec{u} \times \vec{v} \in \mathbb{R}^3$, is completely determined by the following properties:

1. (Anti-symmetry) $\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$
2. (Bilinearity) $\vec{u} \times (\alpha \vec{v} + \beta \vec{w}) = \alpha \vec{u} \times \vec{v} + \beta \vec{u} \times \vec{w}, \forall \alpha, \beta \in \mathbb{R}.$
3. If $\vec{i} = (1, 0, 0), \vec{j} = (0, 1, 0), \vec{k} = (0, 0, 1)$, then $\vec{i} \times \vec{j} = \vec{k}, \vec{j} \times \vec{k} = \vec{i}, \vec{k} \times \vec{i} = \vec{j}.$

Theorem 1. *The following properties hold for any vectors $a, b, c \in \mathbb{R}^3$:*

1. (Signed area interpretation) $|a \times b| = |a||b| \sin \angle(a, b)$, i.e. the length of the cross-product is the area of the parallelogram formed by a and b . In particular, $a \times b = 0 \iff a, b$ are collinear.
2. If $a = (a_1, a_2, a_3), b = (b_1, b_2, b_3)$ then $a \times b = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$
3. (Scalar triple product) $a \cdot (b \times c) = b \cdot (c \times a) = c \cdot (a \times b)$ represents the signed volume of the parallelepiped generated by a, b, c .
4. $a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$
5. (Binet-Cauchy) $(a \times b) \cdot (c \times d) = (a \cdot c)(b \cdot d) - (a \cdot d)(b \cdot c).$

Sometimes a powerful idea is to transform vectors via projections, rotations, reflections, etc. Note that these act *linearly* on vectors.

Theorem 2. *Suppose $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_m$ are vectors in the plane, such that for any line d*

$$|\text{proj}_d(a_1)| + |\text{proj}_d(a_2)| + \dots + |\text{proj}_d(a_n)| \geq |\text{proj}_d(b_1)| + |\text{proj}_d(b_2)| + \dots + |\text{proj}_d(b_m)|.$$

Then

$$|a_1| + |a_2| + \dots + |a_n| \geq |b_1| + |b_2| + \dots + |b_m|.$$

2 Warm-up examples

Example 1. Let ABC be a triangle and $A_1 \in BC$, $B_1 \in AC$, $C_1 \in AB$ be points such that

$$\frac{BA_1}{A_1C} = \frac{CB_1}{B_1A} = \frac{AC_1}{C_1B} = \lambda.$$

Then the segments AA_1, BB_1, CC_1 can form a triangle.

Example 2. Suppose triangles ABC and $A_1B_1C_1$ have the same centroid. Prove that the segments AA_1, BB_1, CC_1 can form a triangle.

Example 3. Let A, B, C, D be distinct points in space. Then

$$1. AC \perp BD \iff AB^2 + CD^2 = AD^2 + BC^2.$$

$$2. \text{ Any 2 of the relations } AB \perp CD, AD \perp BC, AC \perp BD \text{ imply the third.}$$

Solution. We identify points with corresponding position vectors (i.e. $X = \overrightarrow{OX}$ for any point X , where O is a fixed point chosen as an origin). We have

$$\begin{aligned} AB^2 + CD^2 = AD^2 + BC^2 &\iff |A - B|^2 + |C - D|^2 = |A - D|^2 + |B - C|^2 \\ &\iff A \cdot B + C \cdot D = A \cdot D + B \cdot C \iff (A - C) \cdot (B - D) = 0 \iff AC \perp BD. \end{aligned}$$

For the second part of the problem, we have

$$AB \perp CD \iff (A - B) \cdot (C - D) = 0$$

$$AC \perp BD \iff (A - C) \cdot (D - B) = 0$$

$$AD \perp BC \iff (A - D) \cdot (B - C) = 0$$

The sum of the left terms in the above 3 relations is 0. Hence, any two of them imply the third relation. $\square$

Example 4. Let $ABCD$ be a convex quadrilateral with $AB = CD$. If M, N are the midpoints of AD and BC , prove that MN is parallel to the bisector of the angle between the lines AB and CD .

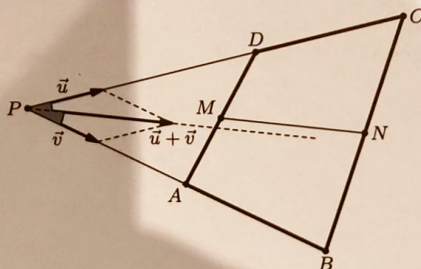


Figure 1: Example 4

Solution. Let $P = AB \cap CD$, and let $\vec{u} = \overrightarrow{PU}$, $\vec{v} = \overrightarrow{PV}$ be unit vectors in the directions of $\overrightarrow{AB}$ and $\overrightarrow{DC}$, respectively. We have

$$\overrightarrow{MN} = \frac{1}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{DC} = a(\vec{u} + \vec{v})$$

where $a = \frac{AB}{2} = \frac{CD}{2}$. It remains to note that, since $|\vec{u}| = |\vec{v}|$, $\vec{u}$ and $\vec{v}$ generate a rhombus, hence $\vec{u} + \vec{v}$ is parallel to the bisector of $\angle DPA$. $\square$

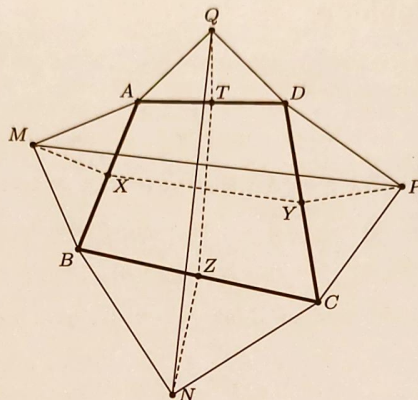


Figure 2: Example 5

Example 5 (IMO Shortlist, 1992). Let $ABCD$ be a convex quadrilateral, and M, N, P, Q be the centers of the squares constructed along sides AB, BC, CD, DA outwards of $ABCD$. Show that $MP \perp NQ$.

Solution. Let X, Y, Z, T be the midpoints of AB, CD, BC, AD . Rotate all vectors anticlockwise by 90° , and denote by $r(\vec{v})$ the image of a vector $\vec{v}$. Then $r(\overrightarrow{MX}) = \overrightarrow{XA} = \frac{1}{2}\overrightarrow{BA}$, and similarly

$$r(\overrightarrow{YP}) = \frac{1}{2}\overrightarrow{CD}, r(\frac{1}{2}\overrightarrow{AD}) = \overrightarrow{TQ}, r(\frac{1}{2}\overrightarrow{BC}) = \overrightarrow{NZ}$$

The transformation r is linear, hence

$$\begin{aligned} r(\overrightarrow{MP}) &= r(\overrightarrow{MX} + \overrightarrow{XY} + \overrightarrow{YP}) = r(\overrightarrow{MX}) + r(\frac{1}{2}\overrightarrow{AD} + \frac{1}{2}\overrightarrow{BC}) + r(\overrightarrow{YP}) = \\ &= \frac{1}{2}\overrightarrow{BA} + \overrightarrow{TQ} + \overrightarrow{NZ} + \frac{1}{2}\overrightarrow{CD} = \overrightarrow{NZ} + \overrightarrow{ZT} + \overrightarrow{TQ} = \overrightarrow{NQ} \end{aligned}$$

Therefore, $MP \perp NQ$ (and $MP = NQ$). $\square$

Example 6. A, B and C move with constant speeds along straight lines. Prove that if the positions of A, B, C are collinear at 3 different instances, then their positions are always collinear.

Solution. The position vector of A at time t has the form $A(t) = u + vt$, for some fixed vectors u, v . It follows that that same is true for vectors AB and AC :

$$\overrightarrow{AB}_t = a + bt, \quad \overrightarrow{AC}_t = c + dt,$$

for some fixed vectors a, b, c, d . Hence

$$\overrightarrow{AB}_t \times \overrightarrow{AC}_t = a \times c + (a \times d + b \times c)t + (b \times d)t^2 = P(t)$$

is a quadratic in $t \in \mathbb{R}$. The positions of A, B, C are collinear at times t_1, t_2, t_3 if and only if $P(t_1) = P(t_2) = P(t_3) = 0$. In this case it follows that the polynomial $P(t)$ is identically 0. Therefore, $\overrightarrow{AB}_t \times \overrightarrow{AC}_t = 0$ always, and the conclusion follows. $\square$

3 More Challenging Problems

Problem 1 (Mobius). If $ABCDE$ is a convex pentagon and S, a, b, c, d, e are the areas of $ABCDE, EAB, ABC, BCD, CDE$ and DEA , respectively, then

$$S^2 - (a + b + c + d + e)S + (ab + bc + cd + de + ea) = 0.$$

Solution. Putting $f = [ACD]$, we have $S = b + f + e$ and the relation to be proved becomes

$$(b + f + e)(b + f + e - a - b - c - d - e) + ab + bc + cd + de + ea = 0 \iff$$

$$f(b + f + e - c) - fc - ec - db - fd + cd = 0 \quad (1)$$

We identify points with their position vectors and take $A = \vec{0}$. We have (making sure all cross products have the same orientation)

$$c = (D - C) \times (B - C) = D \times B + B \times C + C \times D$$

and similarly,

$$a = B \times E, \quad b = B \times C, \quad d = E \times C + C \times D + D \times E, \quad e = D \times E, \quad f = C \times D.$$

With these, one can establish (1) via a direct computation in which terms on the left completely cancel out. $\square$

Problem 2 (IMO 2005). Let ABC be an equilateral triangle and $A_1, A_2 \in BC, B_1, B_2 \in CA, C_1, C_2 \in AB$ such that $A_1A_2B_1B_2C_1C_2$ is a convex hexagon with all sides equal. Prove that A_1B_2, B_1C_2, C_1A_2 concur.

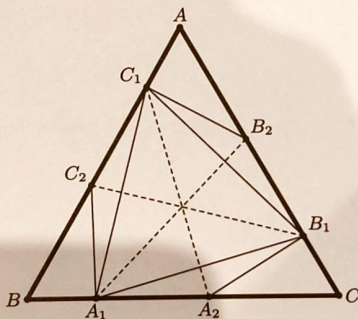


Figure 3: IMO 2005

Proof. We have $\overline{A_1A_2} + \overline{B_1B_2} + \overline{C_1C_2} = 0$, and hence also $\overline{A_2B_1} + \overline{B_2C_1} + \overline{C_2A_1} = 0$. Therefore, these vectors form an equilateral triangle, and in particular are obtained from each other by rotations by 120° (and 240°). The same is true for the vectors $\overline{A_1A_2}, \overline{B_1B_2}, \overline{C_1C_2}$. Therefore, $\triangle C_1C_2A_1 \equiv \triangle A_1A_2B_1 \equiv \triangle B_1B_2C_1$. In particular, $\triangle A_1B_1C_1$ is equilateral, and $B_2B_1 = B_2C_1$ implies that A_1B_2 is a median in $\triangle A_1B_1C_1$. Similarly, B_1C_2, C_1A_2 are also medians in $\triangle A_1B_1C_1$, hence all concur. $\square$

Problem 3 (Kazakhstan 2006). Consider a convex hexagon $ABCDEF$ with

$$AD = BC + EF, \quad BE = CD + FA, \quad CF = AB + DE.$$

Show that

$$\frac{AB}{DE} = \frac{EF}{BC} = \frac{CD}{FA}$$

Solution. Letting $x = \overrightarrow{AB} + \overrightarrow{ED}$, $y = \overrightarrow{CD} + \overrightarrow{FA}$, $z = \overrightarrow{EF} + \overrightarrow{CB}$, we have

$$\overrightarrow{AD} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{AF} + \overrightarrow{FE} + \overrightarrow{ED}) = \frac{x - z + y}{2}$$

and hence, by the problem condition,

$$\left| \frac{x - z + y}{2} \right| = |\overrightarrow{AD}| = BC + EF \geq |\overrightarrow{EF} + \overrightarrow{CB}| = |z|$$

(with equality if $BC \parallel EF$). Similarly,

$$\left| \frac{y - x + z}{2} \right| \geq |x|$$

$$\left| \frac{z - y + x}{2} \right| \geq |y|$$

Now, substituting $u = \frac{y-x+z}{2}$, $v = \frac{z-y+x}{2}$, $w = \frac{x-z+y}{2}$ into the last inequalities we have

$$|u| \geq |v + w|, \quad |v| \geq |w + u|, \quad |w| \geq |u + v|$$

Squaring and adding up these inequalities, we have, after term reductions

$$0 \geq |u|^2 + |v|^2 + |w|^2 + 2(u \cdot v + v \cdot w + w \cdot u) = (u + v + w)^2$$

We must have equalities in all of the above, therefore $u + v + w = \vec{0} = x + y + z$ and the opposite sides of $ABCDEF$ are parallel. Moreover, $\overrightarrow{AD}w = -z = \overrightarrow{FE} + \overrightarrow{BC}$ with $FE \parallel BC$ implies $AD \parallel FE \parallel BC$. Similarly, $AF \parallel BE \parallel CD$. Finally, letting $L = BE \cap CF$, from parallelograms $ABLF$ and $CDEL$ we have

$$\frac{AB}{DE} = \frac{FL}{LC} = \frac{EF}{BC} = \frac{EL}{LB} = \frac{CD}{AF}.$$

□

4 Practice Problems

Problem 4. For any 4 points A, B, C, D in space, if M, N are the midpoints of AC and BD , then

$$AB^2 + BC^2 + CD^2 + DA^2 = AC^2 + BD^2 + 4MN^2$$

Problem 5. If $A_1A_2...A_n$ is a regular polygon with center O , then

$$\overrightarrow{OA_1} + \overrightarrow{OA_2} + \dots + \overrightarrow{OA_n} = \vec{0}.$$

Problem 6. Show that if $u_1, u_2, \dots, u_n$ ($n \geq 3$) are plane non-parallel vectors with $u_1 + u_2 + \dots + u_n = 0$ then there exists a convex polygon $A_1A_2...A_n$ such that $\{\overrightarrow{A_1A_2}, \overrightarrow{A_2A_3}, \dots, \overrightarrow{A_nA_1}\} = \{u_1, u_2, \dots, u_n\}$.

Problem 7. Let $M_1, M_2, \dots, M_6$ be the midpoints of the sides $A_1A_2, A_2A_3, \dots, A_6A_1$ of a convex hexagon.

1. Show that $M_2M_5^2 = M_1M_4^2 + M_3M_6^2 \iff M_1M_4 \perp M_3M_6$.

2. Show that M_1M_4, M_2M_5, M_3M_6 are concurrent $\iff [A_1A_3A_5] = [A_2A_4A_6]$.

Problem 8. Let $ABCD$ be a quadrilateral, $P = AC \cap BD$ and $X, Y \in AB$, $M, N \in CD$ such that $AX = XY = YB$ and $CM = MN = ND$. Show that $[XPM] = [YPN]$.

Problem 9. Does there exist a bijection f of (a) the plane or (b) the 3D-space such that for any distinct points A, B we have $AB \perp f(A)f(B)$?

Problem 10 (BMO 1985). Let O be the circumcenter of $\triangle ABC$, and D be the midpoint of AB . Let E be the centroid of $\triangle ACD$. Prove that $CD \perp OE$ if and only if $AB = AC$.

Problem 11 (IMO 1990 proposal). Given $\triangle ABC$ with distinct side lengths, let G , I , and H be its centroid, incenter, and orthocenter, respectively. Prove that $\angle GIH > 90^\circ$.

Problem 12. Let $A_1A_2A_3A_4A_5A_6$ be a convex hexagon with opposite sides parallel. Prove that triangles $A_1A_3A_5$ and $A_2A_4A_6$ have equal areas.

Problem 13 (Crux Problem 2333). Let D and E be points on sides AC and AB of $\triangle ABC$, respectively, with $DE \nparallel CB$. Suppose F and G lie on BC and ED such that:

$$\frac{BF}{FC} = \frac{EG}{GD} = \frac{BE}{CD}.$$

Show that GF is parallel to the angle bisector of $\angle BAC$.

Problem 14. Let $ABCD$ be a cyclic quadrilateral, $E = AD \cap BC$ and $F = AB \cap CD$. Show that the bisectors of $\angle AEB$ and $\angle AFD$ are perpendicular.

Problem 15. Let $ABCDE$ be a convex pentagon.

- (BMO 1996) If A_1, B_1, C_1, D_1, E_1 are the midpoints of CD, DE, EA, AB and BC , respectively, and $X \in AA_1 \cap BB_1 \cap CC_1 \cap DD_1$, then $X \in EE_1$.
- Any 4 of the conditions $AB \parallel CE, BC \parallel DA, CD \parallel EB, DE \parallel AC, EA \parallel BD$ imply the fifth.

Problem 16 (USSR 1976). Given any vectors a, b, c show that

$$|a| + |b| + |c| + |a + b + c| \geq |a + b| + |a + c| + |b + c|$$

Problem 17. Let H_1, H_2, H_3 be the orthocenters of triangles $A_2A_3A_4, A_1A_3A_4$ and $A_1A_2A_4$. Prove that $[A_1A_2A_3] = [H_1H_2H_3]$.

Problem 18. If convex polygon $B_1B_2 \dots B_n$ is inside polygon $A_1A_2 \dots A_n$, then the perimeter of $B_1B_2 \dots B_n$ does not exceed the perimeter of $A_1A_2 \dots A_n$.

Problem 19. Orthogonal to each face of a polyhedron construct an outward vector with length numerically equal to the area of the face. Prove that the sum of all these vectors is equal to zero.

(It's enough to show the claim in the case of a tetrahedron).

Problem 20 (Vietnam TST 2004). Let $ABCDEF$ be a convex hexagon with perimeter P , and the midpoints of its sides form the hexagon $A_1B_1C_1D_1E_1F_1$ with perimeter P' . If all angles of $A_1B_1C_1D_1E_1F_1$ are equal, prove that $P \geq \frac{2\sqrt{3}}{3}P'$.

Problem 21 (IMO 2003). In convex hexagon $ABCDEF$, the distance between the midpoints of any pair of opposite sides equals $\frac{\sqrt{3}}{2}$ times the sum of their lengths. Show that all angles of the hexagon are equal.